

Aircraft Tail Aerodynamic Configuration Variables

Concise reference of tail variables that can be changed to affect aerodynamics and stability.

1. Horizontal tail geometry

Variable / configuration	Primary aerodynamic effect
Horizontal tail area	Affects pitch control authority and longitudinal stability.
Horizontal tail span	Changes tail efficiency and pitch-control authority.
Horizontal tail aspect ratio	Higher AR generally reduces induced drag and improves efficiency.
Horizontal tail taper ratio	Changes lift distribution and stall behaviour.
Horizontal tail sweep	Affects compressibility, stability and stall characteristics.
Horizontal tail thickness	Affects drag and structural characteristics.
Horizontal tail airfoil	Changes lift, drag and pitching-moment characteristics.
Horizontal tail camber	Changes tail lift and pitching moment.
Horizontal tail twist	Changes spanwise loading and stall behaviour.
Horizontal tail incidence	Changes trim and required elevator deflection.
Horizontal tail dihedral/anedral	Changes lateral and directional coupling.
Horizontal tail longitudinal position	Changes moment arm and pitch authority.
Horizontal tail vertical position	Changes interaction with wing and fuselage flow.

2. Elevator

Variable / configuration	Primary aerodynamic effect
Elevator area	Changes pitch-control authority.
Elevator chord	Changes pitch authority, drag and hinge moment.
Elevator span	Changes total pitch-control authority.
Elevator deflection	Changes pitching moment and drag.
Elevator hinge position	Changes control effectiveness and hinge moment.
Elevator balance	Changes aerodynamic/control forces and required actuator effort.

3. Vertical tail geometry

Variable / configuration	Primary aerodynamic effect
Vertical tail area	Affects directional stability and yaw-control authority.
Vertical tail height	Changes directional stability and yaw moment.
Vertical tail aspect ratio	Higher AR generally improves efficiency and reduces induced drag.
Vertical tail taper ratio	Changes loading and aerodynamic efficiency.
Vertical tail sweep	Affects high-speed behaviour and directional stability.
Vertical tail thickness	Affects drag and structural characteristics.
Vertical-tail airfoil	Changes side-force, drag and moment characteristics.
Vertical-tail camber	Changes side-force characteristics and trim.
Vertical-tail incidence	Changes side-force/trim behaviour.
Vertical-tail longitudinal position	Changes yaw moment arm and effectiveness.

4. Rudder

Variable / configuration	Primary aerodynamic effect
Rudder area	Changes yaw-control authority.
Rudder chord	Changes yaw authority, drag and hinge moment.
Rudder span/height	Changes total yaw-control authority.
Rudder deflection	Changes yawing moment and drag.
Rudder hinge position	Changes control effectiveness and loads.

5. Tail configuration/layout

Variable / configuration	Primary aerodynamic effect
Conventional tail	Horizontal tail behind fuselage; simple and common configuration.
T-tail	Moves horizontal tail above the fuselage wake; changes stability and stall/interference behaviour.
H-tail	Twin vertical tails linked by a horizontal tail; can provide redundancy and altered interference.
V-tail	Combines pitch and yaw functions into two inclined surfaces; reduces surface count but couples controls.
Y-tail	Three-surface arrangement combining aspects of conventional/V-tail layouts.

Variable / configuration	Primary aerodynamic effect
Twin-tail	Two vertical stabilisers; can improve redundancy, control authority and packaging.
Twin-boom tail	Tail surfaces mounted between/behind booms; changes interference and moment arm.
Canard	Forward lifting/control surface; changes trim, stability and stall characteristics.
Three-surface configuration	Canard + main wing + tail; distributes lift and pitching moments.
Boom-mounted tail	Separates tail from fuselage and changes aerodynamic interference.

6. Tail placement and integration

Variable / configuration	Primary aerodynamic effect
Horizontal-tail longitudinal position	Changes pitch moment arm and required tail force.
Horizontal-tail vertical position	Changes wing/tail interference and flow quality.
Vertical-tail longitudinal position	Changes yaw moment arm and control effectiveness.
Vertical-tail height	Changes directional stability and exposure to clean airflow.
Tail incidence	Changes trim and control-surface requirements.
Tail-fuselage junction geometry	Can reduce or increase interference drag and separation.
Tail-wing separation	Changes aerodynamic interaction between surfaces.

7. Canard variables

Variable / configuration	Primary aerodynamic effect
Canard area	Affects pitch authority, trim and lift distribution.
Canard span	Changes efficiency and pitch-control authority.
Canard aspect ratio	Higher AR generally reduces induced drag.
Canard sweep	Affects high-speed behaviour and stall characteristics.
Canard taper	Changes lift distribution and stall progression.
Canard thickness	Affects drag and structural volume.
Canard airfoil	Controls lift, drag and pitching-moment characteristics.
Canard camber	Changes lift and pitching moment.
Canard incidence	Changes trim and lift sharing with the main wing.
Canard twist	Changes spanwise loading and stall behaviour.
Canard longitudinal position	Changes pitch moment arm and stability.
Canard vertical position	Changes interference with the fuselage/main wing.
Canard elevator/control-surface size	Changes pitch-control authority.
Canard deflection	Changes pitching moment and drag.